

Multimodal Emotion Recognition in Noisy Environment Based on Progressive Label Revision

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ABSTRACT

The multimodal emotion recognition has attracted more attention in recent decades. Though remarkable progress has been achieved with the rapid development of deep learning, existing methods are still hard to tackle noise problems that occurred commonly in emotion recognition's practical application. To improve the robustness of the multimodal emotion recognition algorithm, we propose an MLP-based label revision algorithm. The framework consists of three complementary feature extraction networks that were verified in MER2023. After that, an MLP-based attention network with specially designed loss functions was used to fuse features from different modalities. Finally, the scheme that used the output probability of each emotion to revise the sample's output category was employed to revise the test set's label obtained by classifier. The samples that are most likely to be affected by noise and misclassified have a chance to get correct classification. The best experimental result shows that the F1-score of our algorithm on the test dataset

of the MER 2023 Noise subchallenge is 86.35 and combined metric is 0.6694, which ranks 2nd at the MER 2023 NOISE subchallenge.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**;

KEYWORDS

Multimodal emotion recognition, neural networks, modality robustness, feature fusion, label revision

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1 INTRODUCTION

Diverse emotion recognition applications have been rapidly developing in human-computer interaction and have attracted extensive attention from researchers [17, 18]. During past decades, conventional statistical-based methods have always been one of the mainstream approaches for multimodal emotion recognition. For example, Kaya et. al. [11] employed least square-based learners to tackle traditional features to complete an emotion recognition modal. In

2015, Wu et al. [30] encoded video clips to features by series models. And many literature focuses on a variety of hand-crafted features like Local Binary Patterns to represent emotional information [10, 32], then these features are fed into machine learning-based classifiers such as SVM etc.[26].

Despite of success already achieved, the statistical-based algorithms are not robust enough to meet the requirements of applications in complex real-life scenarios. To alleviate the model robustness, with the rapid development of deep learning, researchers resort to directly applying deep neural networks, such as Alexnet [14], GoogLeNet [24], VGGnet [21], ResNet [8], to obtain more stable models. Thus, based on the success of neural networks in other fields, deep neural networks have rapidly taken over the mainstream in emotion recognition in complex scenes, e.g., speaker-independent [19], cross-corpus emotion recognition [34, 35], and emotion recognition in the wild [11]. In [27], Wang et. al proposed Self-Cure-Network that relabels samples with the confidence level for reducing the influence of the label's uncertainties. Inspired by this work, in this paper, we propose an attention-based progressive label revise network (PLRN) that based on the output probability, samples with the lowest confidence level are relabeled with other emotion classes.

The remainder of this paper is organized as follows. The framework of our modal including feature extraction, feature fusion, and label revision is described in section 2. In section 3, we compare our experimental results with the baseline on the challenge dataset to evaluate the effectiveness of our proposed method. Finally, in section 4, we conclude this paper.

2 THE PROPOSED METHOD

Our proposed framework consists of three parts, i.e. feature extraction block, deep learning-based emotion feature extraction block, and label revision block. The feature extraction block consists of three well-performed DNN-based networks for unique modal feature learning. As for the emotion feature extraction block, there are three MLP-based deep neural networks and an attention-based fusion module. And the label revision block is a self-cure network that revises the test sample's label with the lowest predicted probability.

2.1 Feature Extraction and Fusion

The videos in first split into audios and videos, then videos were fed into the OpenFace toolkit which was developed by Tadas Baltrušaitis et. al in 2016 [1]. After Openface the face in the video are cropped and aligned into 96*96. Then ResNet [8] pretrained on the FER2013 dataset [7] is used to extract emotion features from cropped facial images. In text-related tasks, BERT [12] and its variants have achieved great success with the use of the 'masked language model (MLM)' and 'next sentence prediction (NSP)' to learn contextual word embeddings. Considering the fact that all videos are collected from China, we choose the MacBert which are proposed by Yiming Cui et. al in 2022 as the feature extractor [4]. As for the audio modal, we used Hubert proposed by Wei-Ning Hsu et. al in 2021 [9] that are developed based on the BERT. The Hubert achieved better performance than widely used wav2vec [2] on most tasks including the MER challenge.

After the generalization feature extracted by methods mentioned above, we use three parallel MLPs as emotion feature extractors, each MLP consists of several stacked fully connected layers that perform linear transformation on input to get unimodal feature $f_i^m \in R^{d_m}$ and using ReLU as activation function to get each modal's hidden features $h_i^m \in R^h$ as described as:

$$h_i^m = \text{ReLU}(f_i^m W_m^h + b_m^h), m \in \{a, l, v\}. \quad (1)$$

However, due to the fact that the test set is corrupted by noise, it's hard to get a satisfactory performance with the original training set. So we first get the contaminated training set as described in [16, 22] and get corresponding hidden features hn_i^m as:

$$hn_i^m = \text{ReLU}(f_i^m W_m^h + b_m^h), m \in \{a, l\}. \quad (2)$$

then the residual function calculated by hidden feature from original audio h_i^m and noise audio hn_i^m been introduced to reduce the gap between original feature and noise feature:

$$L_{res} = \text{sum}(\|h_i^m - hn_i^m\|), m \in \{a, l\} \quad (3)$$

By the introduction of residual function, feature extractor can learn to extract features that are more robust to noise.

For fusing different modal features, we first stack these features as h_i , then use the attention network to calculate the importance of each modal $a \in R^{3 \times 1}$. Then we obtain weighted fusion multi modal features $z = \sum_i a_i h_i$. The specific calculation process can be presented as:

$$h_i = \text{Concat}(h_i^a, h_i^l, h_i^v) \quad (4)$$

$$a_i = \text{softmax}(h_i^T W_a + b_a) \quad (5)$$

Considering the experiment result in [16] that the most severely affected modal of performance by noise is audio. Unfortunately, the audio is superior to the other two modals in performance without noise, which means that audio's attention weight tends to be relatively high. So the introduction of a loss function on weight a_i to prevent overreliance on audio is helpful. So we introduce margin loss:

$$L_{att} = \max(0, \text{margin}(a_H - a_L)) \quad (6)$$

where a_H and a_L represent the maximum and minimum of attention weight. In order for this residual function to balance the weights between different modes as much as possible without prior knowledge, the margin is set to be very small: 0.03 rather than a larger value. But on the other hand, considering that this margin loss function does not directly affect the performance, in order to avoid overfitting the residual function to the extent that the attention weights are basically equal, we multiply a coefficient in front of it to ensure that it is significantly smaller than the cross entropy loss function.

After concatenation, the hidden feature is linearly transformed to predict the valence according to Eq. (7) and the mean squared error (MSE) in Eq. (8) is used as the regression loss to guide the training procedure:

$$\hat{v}_i = h_i^m W_m^v + b_m^v, m \in \{a, l, v\} \quad (7)$$

$$L_v = \frac{1}{N_l} \sum_{i=1}^{N_l} (v_i - \hat{v}_i)^2 \quad (8)$$

Similarly, the predicted result of classification is trained and obtained by Eq. (9)(10).

$$\hat{e}_i = \text{soft max}(h_i^m W_m^e + b_m^e), m \in \{a, l, v\} \quad (9)$$

$$L_e = -\frac{1}{N_l} \sum_{i=1}^{N_l} e_i \log(\hat{e}_i) \quad (10)$$

During training procedure, $W_m^h \in R^{d_m * h}$, $b_m^h \in R^h$, $W_m^v \in R^{d_m * 1}$, $b_m^v \in R^1$, $W_m^e \in R^{d_m * C}$ and $b_m^e \in R^C$ are trainable parameters. To avoid mutual interference between the classification task and valence regression task, we split the training procedure into two: first, we only train the regression part using $L_1 = L_v$. Second, we use the combination of different loss functions: $L_2 = L_e + \lambda_1 L_{res} + \lambda_2 L_{att}$ to train classification result. The parameter λ_1 and λ_2 are set at $1e-6$ and $1e-3$. And the overall feature extraction network can be summarized as progressive label revise feature extraction network (PLRFEN), and the structure can be seen in Fig 1(a).

2.2 Label Revise Strategy

In previous work, we present the procedure of emotion feature extraction of different modalities, the fusion of these features, and the output of final probability. Due to the fact that the best F1-score we get by the method described above is about 0.7, it's easy to speculate that there are quite a few incorrect classification results in the output. Inspired by the work [27], we can infer that samples most difficult to fit and have the lowest confidence probability are most likely to be classified incorrectly. Especially considering the fact that in the multimodal emotion recognition task in noisy environments, different modalities may indeed correspond to different emotions rather than the same emotion. And these samples often have lower prediction probabilities due to modal conflicts. So after a better classification result obtained by average on nine different classification results, we add the test set of MER-NOISE into the training set as an auxiliary dataset and this result is used as the initial pseudo label. Afterward, when we train our network and use it to get the predicted output of the test set: $[p_1^1, \dots, p_1^c] \dots [p_n^1, \dots, p_n^c]$, where c represent the number of emotion categories and n represent the number of samples in test dataset. Then each sample is labeled by emotion corresponding highest probability: $p_k^{\max} = \max(p_k^1, \dots, p_k^c)$. Then we choose the several samples with the lowest probability and corresponding result $p_k^{\max} = \min(p_1^{\max}, \dots, p_n^{\max})$, we will relabel these samples by the emotion category that has the second highest predicted probability $p_n^{\max-1}$. When the training process is completed, the epoch with results that can fit the training set and has lower loss of confidence in relabel procedure of the test set ($accuracy_{train} > \sigma$ and $\sum_k (p_k^{\max} - p_k^{\max-1}) < \theta$) is selected as the final submission result. Due to the fact that labels of training set are real, which means they should be easy to fit on the trained network the value of σ will be set to close to 1. As for the value of θ , to

avoid having no choice, we usually set the value to a larger one and choose the epoch corresponding to the smallest $\sum_k (p_k^{\max} - p_k^{\max-1})$.

The whole procedure of our PLRN can be seen in Fig 1(b).

2.3 Implementation Details

There is mainly the modification of hyper-parameter in our method compared with the baseline paper. The hidden feature size at different stages was set at 128 and 256 respectively in the training procedure. And during optimization procedure, Adam [13] is selected as the optimizer, and the learning rate is set at $3e-4$ according to experiment performance rather than $1e-3, 1e-4, 1e-5$ used in the baseline paper. And due to the series overfitting problem in the noise subchallenge, the dropout [23] ratio is increased from 0.3 to 0.5. The maximum number of epochs and weight decay are set at 100 and $1e-5$ respectively.

3 EXPERIMENT

3.1 Dataset and Preprocessing

The training set and test set used in MER NOISE subchallenge are the extension of CHEAVD [15]. Finally, 3373 samples are selected as training set and 412 samples are selected as test set [16]. Each sample are annotated by at least three annotators and use majority voting as final label. Then a agreement check and reliability check have been conducted on the annotation to ensure annotation accuracy [6, 20, 25, 28, 29, 31, 33]. After emotion class has been annotated, six annotator have been assigned to label valence between $[-5, 5]$ using popular self-assessment manikin [3]. To ensure annotation quality, highest and lowest value has been excluded and average of remaining 4 values are adapted as final valence value. For evaluating methods in noisy environment, randomly selected noises from MUSAN(noise corpus containing three subsets music, noise, and speech)[22] speech subset are added to the test video and speech-to-noise ratio (SNR) is set between 5-10. And the visual modality is blurred by downsample and upsample to get low-resolution images.

For mitigation the noise's influence, we first add the noise in MUSAN to training set to reduce the difference between training set and test set. Furthermore, we use denoiser toolkit to process test set and get a cleaner dataset. The denoiser toolkit is proposed in Interspeech 2020 by Alexandre Defossez et.al in [5]. The denoiser process original waveform using causal Speech enhancement model and series loss functions. The result show it can suppress background noise to some extent. As for the video blurring, considering the openface used in our method can extract face satisfactory even in test set and the result in [16] show the performance of video modality wouldn't deteriorate. We don't take any additional measures for video modality.

3.2 Result and Discussion

In the MER 2023 NOISE subchallenge, we adopt proposed progressive label revise network to predict test set's label and the result are evaluate on two different aspect. For classification result, due to the class imbalance of used dataset, F1-score are used as metric. For valence regression, MSE regression loss is used as metric. Finally combined metric $metric = metric_{cls} - 0.25 * metric_{MSE}$ are used for ranking different result and out best result achieve

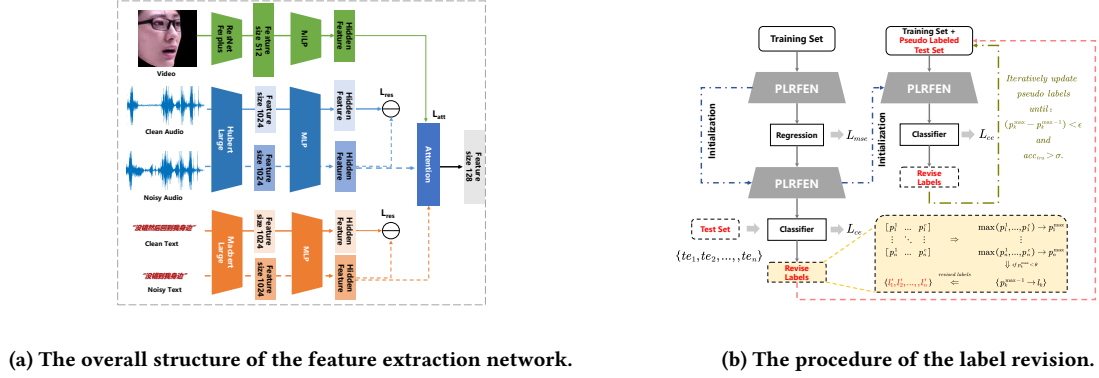


Figure 1: The overall structure of proposed progressive label revise network.

Table 1: Comparison results in terms of Classification / F1-Score, Valence Dimension/ MSELoss and Combined metrics between different modals

Method	Classification	Valence	Combined metrics
Hubert-large	57.14±0.39	1.63±0.03	0.16±0.01
Macbert-large	42.36±0.49	2.01±0.02	-0.08±0.00
ResNet-ferplus	57.68±0.38	1.39±0.02	0.23±0.01
Hubert-large+Macbert-large+ResNet-ferplus	68.35±0.54	1.20±0.04	0.38±0.01
Hubert-large+Macbert-large+ResNet-ferplus with noise trainingset	68.59	0.8137	0.4825
PLRFEN	71.65	0.7763	0.5224
PLRFEN voting average	78.10	0.7763	0.5869
PLRN	86.35	0.7763	0.6694

86.35 on F1-score, 0.7763 on MSE of valence regression and 0.6694 on combined metric. The result of our networks and comparison methods are all listed in Table 1. In Table 1, the result of Hubert-large, Macbert-large, ResNet-ferplus and Hubert-large+Macbert-large+ResNet-ferplus are all get from baseline paper [16]. The result of Hubert-large+Macbert-large+ResNet-ferplus with noise trainingset represent recurrent of baseling paper with noise contaminate training set, which means the reducing of gap between training set and test set can improve performance significantly. However, the improvment mainly focused on the valence regression which means the features contributed to emotion classification still worse than we need. Furthermore, using the PLRFEN that have residual loss function and attention loss fuction we proposed in this paper, we futher force the network to exracted emotion related feature whether there have noise or not, and the F1-score improved from 68.59 to 71.65 and combined metric further increased from 0.4825 to 0.5524 show the effectiveness of our proposed loss function. Considering the fact that 71.65 on F1-score is still lower than we need that there are still about 30 percent samples been misclassified. For providing a more accurate starting to our label revise network, we conduct PLRFEN nine times and average their results to get 'PLRFEN voting average' result that improve F1-score significantly from 71.65 to 78.10. Due to the experiment result that there are no improvement on valen regression using average result, we still using previous result. Finally, we use the result of 'PLRFEN voting average' as the start of PLRN, the classification result was a gradual improvement.

After several iterations of training, testing and updating of auxiliary sets, the best F1-score result we have achieved is 86.35 and the final combined result is 0.6694.

4 CONCLUSION

In this paper, we present a progressive label revise network for noise to recognize the emotion of video corrupted by noise and blur. Emotion information is extracted from three complementary modalities: audio, video, and text with generally used network. Then we use MLP based scheme to extract and fuse emotional features. Finally, a relabel method was used to revise labels corresponsding samples have lowest confidence level in the test dataset, which we speculate that noise affect different modal in different degree and result in conflicting discriminative information for different modalities. The experiment result on the subchallenge test set showed that our proposed method is more robust to the noise met in the emotion recognition task.

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